

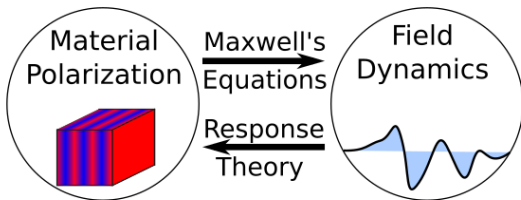
Response Theory

Mike Reppert

October 2, 2026

Previously on CHM676...

In homogeneous dielectric materials, the dynamics of $\mathbf{E}$ and $\mathbf{B}$ are determined by the polarization density $\mathbf{P}$.



Today: How does $\mathbf{P}$ respond to the field?

Outline for Today:

- 1 Physical Guidelines
- 2 Mathematical Framework
- 3 Symmetry and Invariance of Response Tensors

Physical Guidelines

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We study $\boldsymbol{P}$ as a *functional* of $\boldsymbol{E}$ and $\boldsymbol{B}$:

$$\boldsymbol{P}(\boldsymbol{x}, t) = \boldsymbol{P}[\boldsymbol{E}(\boldsymbol{x}', t'), \boldsymbol{B}(\boldsymbol{x}', t')].$$

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 - Must exist a time scale δt *below* which P no longer cares about variations in $E(t + \delta t)$
 - Must exist a time scale T *beyond* which P doesn't remember $E(t - T)$.

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Take-Home Points

Physical constraints:

- locality
- causality
- stability

strongly limit the possible forms for the **mathematical** dependence of P on E .

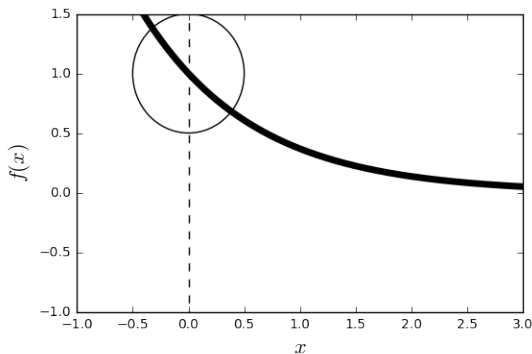
Mathematical Framework

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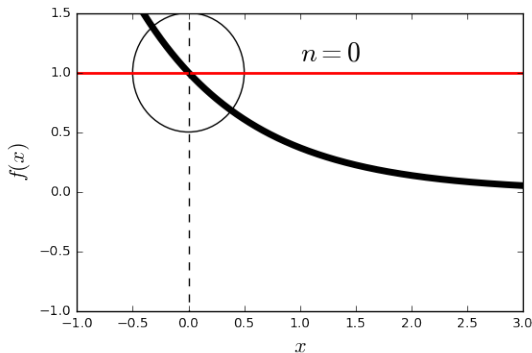
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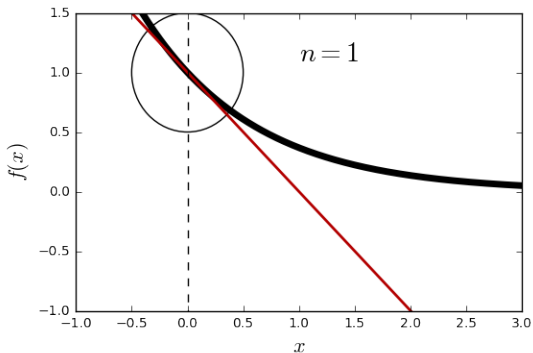
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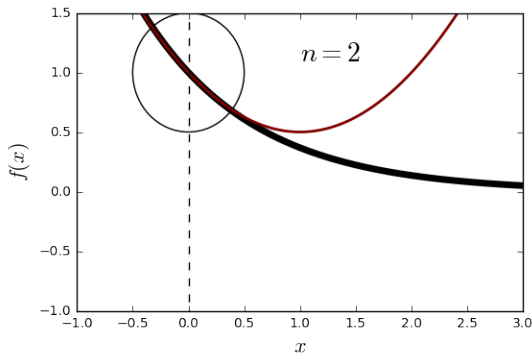
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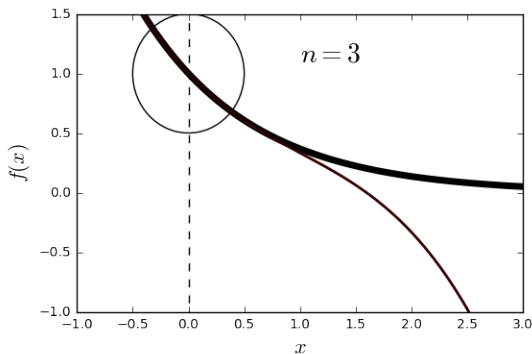
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$$f(x) \approx f(0) + \left. \frac{df}{dx} \right|_{x=0} x + \frac{1}{2} \left. \frac{d^2 f}{dx^2} \right|_{x=0} x^2$$

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Mathematical Framework

The Taylor series of a multi-variable function $g(x_1, \dots, x_N)$ looks like:

$$\begin{aligned}
 g(x_1, \dots, x_N) = & g(0, \dots, 0) \\
 & + \frac{\partial g}{\partial x_1} \Big|_{\mathbf{x}=\mathbf{0}} x_1 + \frac{\partial g}{\partial x_2} \Big|_{\mathbf{x}=\mathbf{0}} x_2 + \dots + \frac{\partial g}{\partial x_N} \Big|_{\mathbf{x}=\mathbf{0}} x_N \\
 & + \frac{1}{2!} \frac{\partial^2 g}{\partial x_1^2} \Big|_{\mathbf{x}=\mathbf{0}} x_1^2 + \frac{\partial g}{\partial x_1 \partial x_2} \Big|_{\mathbf{x}=\mathbf{0}} x_1 x_2 + \dots + \frac{1}{2!} \frac{\partial^2 g}{\partial x_N^2} \Big|_{\mathbf{x}=\mathbf{0}} x_N^2 \\
 & + \dots
 \end{aligned}$$

What is the corresponding expansion for a functional like $P[\mathbf{E}]$?

Mathematical Framework

Since the response is stable, we can sample $\mathbf{E}$ at a finite number of time points:

$$P_I(t) \approx f_I(E_x(t_0), E_y(t_0), E_z(t_0), E_x(t_1), \dots, E_z(t_N); t, \delta t, T).$$

Expanding in a Taylor series:

$$\begin{aligned} P_I(t) &\approx f_I(0, \dots, 0; t, \delta t, T) \\ &+ \frac{\partial f_I}{\partial E_x(t_0)} \Big|_{\mathbf{E}=0} E_x(t_0) + \dots + \frac{\partial f_I}{\partial E_z(t_N)} \Big|_{\mathbf{E}=0} E_z(t_N) \\ &+ \frac{1}{2!} \frac{\partial^2 f_I}{\partial [E_x(t_0)]^2} \Big|_{\mathbf{E}=0} [E_x(t_0)]^2 + \frac{\partial^2 f_I}{\partial E_x(t_0) \partial E_y(t_0)} \Big|_{\mathbf{E}=0} E_x(t_0) E_y(t_0) + \dots \end{aligned}$$

Mathematical Framework

As our sampling points get closer together, the sums converge to integrals:

$$\begin{aligned}
 P(t) = & \sum_{n=0}^{\infty} \sum_{\alpha_1, \dots, \alpha_n} \int_{-\infty}^t dt_n \int_{-\infty}^{t_n} dt_{n-1} \dots \int_{-\infty}^{t_2} dt_1 \\
 & \times E_{\alpha_1}(t_1) E_{\alpha_2}(t_2) \dots E_{\alpha_n}(t_n) \\
 & \times R_{\alpha_1 \dots \alpha_n \alpha}^{(n)}(t, t_n, t_{n-1}, \dots, t_1)
 \end{aligned}$$

where $R_{\alpha_1 \dots \alpha_n \alpha}^{(n)}(t, t_n, t_{n-1}, \dots, t_1)$ is the n^{th} -order *response function** – the target of n^{th} -order spectroscopies.

*Almost. Actually $R^{(n)}$ depends only on time *differences*. Stay tuned!

Symmetry and Invariance of Response Tensors

Time-translation Invariance

All systems we study will satisfy **time-translation invariance**: Only time *differences* matter!

$$R_{\alpha_1 \dots \alpha_n \alpha}^{(n)}(t, t_n, t_{n-1}, \dots, t_1) \Rightarrow R_{\alpha_1 \dots \alpha_n \alpha}^{(n)}(t - t_n, t_n - t_{n-1}, \dots, t_2 - t_1)$$

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Rearranging:

$$P_{\alpha}^{(n)}(t) = \sum_{\alpha_1, \dots, \alpha_n} \int_{-\infty}^{\infty} d\tau_n \dots \int_{-\infty}^{\infty} d\tau_1 R_{\alpha_1 \dots \alpha_n \alpha}^{(n)}(\tau_1, \dots, \tau_n) \\ \times E_{\alpha_1}(t - \tau_1 - \dots - \tau_n) E_{\alpha_2}(t - \tau_2 - \dots - \tau_n) \dots E_{\alpha_n}(t - \tau_n).$$

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Causality dictates that $R_{\alpha_1, \dots, \alpha_n, \alpha}^{(n)}$ is non-zero only for *positive time delays*.

Spatial Symmetries

Neumann's Principle: Spatial symmetries of the material *must* be reflected in the response tensor.

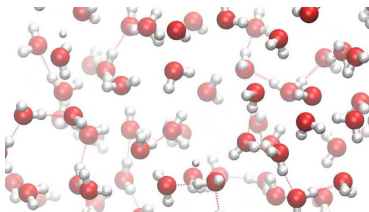
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NB: Only *macroscopic* symmetry is relevant!

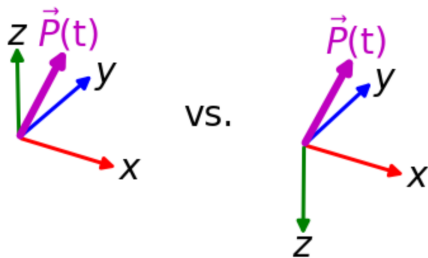


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Spatial Symmetries

Example: $R_{xz}^{(1)}$ in an isotropic (and achiral) sample

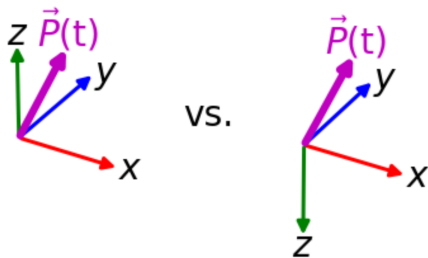
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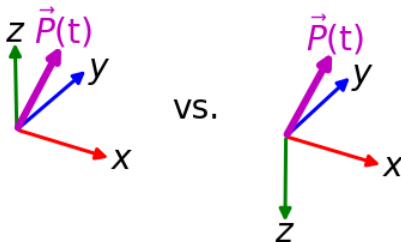


It gets inverted: $P_z \rightarrow -P_z$

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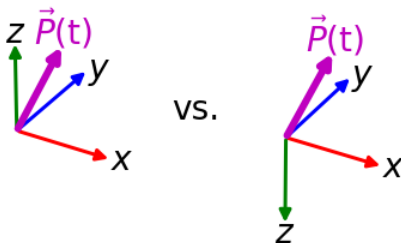
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What happens to $P_x^{(1)}$ when we invert the z -axis?



Nothing! $P_x \rightarrow P_x$

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Under z -axis inversion:

- $z \rightarrow -z$
- $P_x^{(1)} \rightarrow P_x^{(1)}$
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But response theory says:

$$\begin{aligned}
 P_x^{(1)}(t) &= \int_{-\infty}^{\infty} d\tau_1 R_{xz}^{(1)}(\tau_1) E_z(t - \tau_1) \\
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The only possible conclusion is that $R_{xy}^{(1)} = 0!$

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More generally: In isotropic, achiral media

- All tensor elements with an odd number *of any index* vanish (e.g., $R_{xxxy}^{(3)} = 0$)

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More generally: In isotropic, achiral media

- All tensor elements with an odd number *of any index* vanish (e.g., $R_{xxxy}^{(3)} = 0$)
- Corollary: all even-order response functions vanish(!)
- Response tensor elements are symmetry-related (e.g., $R_{xxyy}^{(3)} = R_{yyxx}^{(1)}$)

Even-Order Spectroscopies

But remember: Not all materials are isotropic!

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⇒ Imaging!

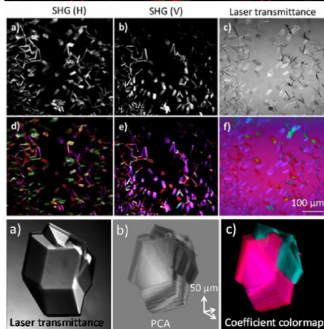
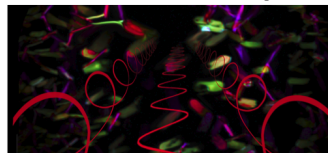
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 $\Rightarrow$ Imaging!



Garth Simpson



Chiral Materials

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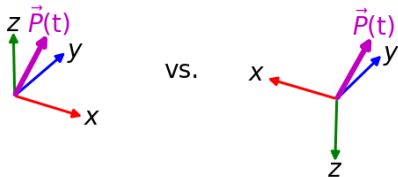
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In a chiral system, inverting any axis flips chirality: If enantiomers are not balanced, this is no longer the same medium, and Neumann's principle does not apply!

But *almost* the same rule is obtained by a 180° rotation:



Here

- $x \rightarrow -x$
- $y \rightarrow y$
- $z \rightarrow -z$

Symmetry shows that $R_{yz} = 0$ (likewise R_{xz} , R_{xy} , etc.)

But (potentially): $R_{xyz} \neq 0$!

In general in a *chiral*, isotropic medium: Any response tensor with any unpaired index vanishes unless x , y , and z are **all** unpaired.

Take-Home Points

Time-translation invariance and **causality** dictate that response functions depend only on *positive time delays* between interactions.

Spatial symmetries in the material must be reflected in the response tensors.

In **achiral isotropic media**:

- Response elements with unpaired axes vanish
- Surviving elements are symmetry-related
- Even-order spectroscopies are forbidden – hence useful for detecting defects